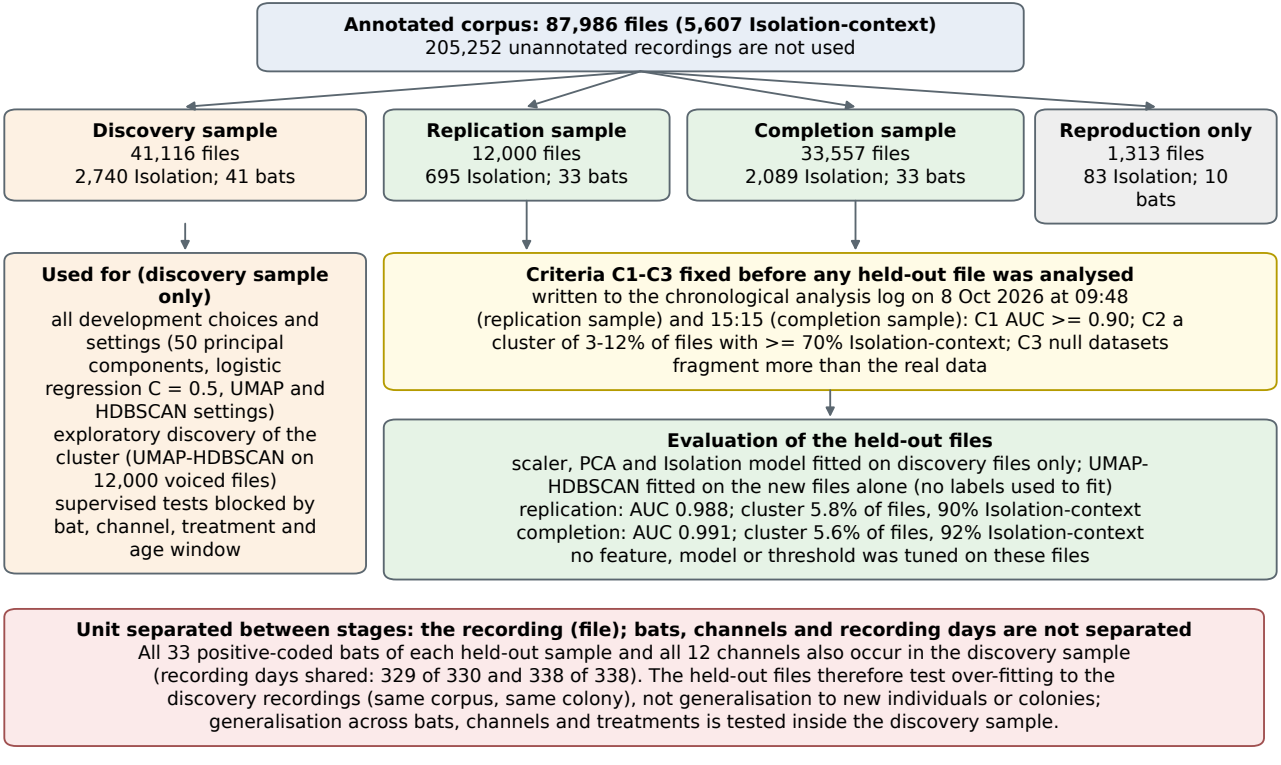
Annotated corpus: 87,986 files (5,607 Isolation-context)
205,252 unannotated recordings are not used
Discovery sample
41,116 files
2,740 Isolation; 41 bats
Replication sample
12,000 files
695 Isolation; 33 bats
Completion sample
33,557 files
2,089 Isolation; 33 bats
Reproduction only
1,313 files
83 Isolation; 10 bats
Used for (discovery sample only)
all development choices and settings (50 principal components, logistic regression C = 0.5, UMAP and HDBSCAN settings)
exploratory discovery of the cluster (UMAP-HDBSCAN on 12,000 voiced files)
supervised tests blocked by bat, channel, treatment and age window
Criteria C1-C3 fixed before any held-out file was analysed
written to the chronological analysis log on 8 Oct 2026 at 09:48 (replication sample) and 15:15 (completion sample): C1 AUC >= 0.90; C2 a cluster of 3-12% of files with >= 70% Isolation-context; C3 null datasets fragment more than the real data
Evaluation of the held-out files
scaler, PCA and Isolation model fitted on discovery files only; UMAP-HDBSCAN fitted on the new files alone (no labels used to fit)
replication: AUC 0.988; cluster 5.8% of files, 90% Isolation-context
completion: AUC 0.991; cluster 5.6% of files, 92% Isolation-context
no feature, model or threshold was tuned on these files
Unit separated between stages: the recording (file); bats, channels and recording days are not separated
All 33 positive-coded bats of each held-out sample and all 12 channels also occur in the discovery sample (recording days shared: 329 of 330 and 338 of 338). The held-out files therefore test over-fitting to the discovery recordings (same corpus, same colony), not generalisation to new individuals or colonies; generalisation across bats, channels and treatments is tested inside the discovery sample.